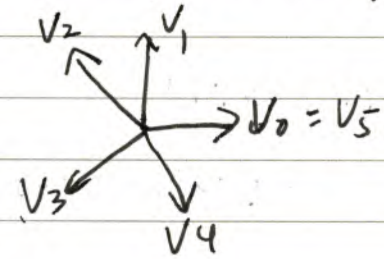
Date

Nonsingular surface

Consider 2-d nonsingular complete toric varieties. $X(\Delta)$

$N = \mathbb{Z}^2$, $\Delta \subseteq N_{\mathbb{R}}$ is given by a sequence of lattice points $v_0, v_1, \cdots, v_d = v_0$



- complete $\sim$ $|\Delta| = N_{\mathbb{R}}$
- nonsigular $\sim$ $\forall\, 0 \le i \le d-1$ V_i and V_{i+1} are a basis for $N \otimes \mathbb{R} = \mathbb{R}^2$

$\Rightarrow \begin{pmatrix} 0 & 1 \\ -1 & a_1 \end{pmatrix} \begin{pmatrix} V_0 \\ V_1 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$ i.e $V_2 = -V_0 + a_1 V_1$

$\Rightarrow a_i V_i = V_{i-1} + V_{i+1}$ $\forall\, 1 \le i \le d$, $a_i \in \mathbb{Z}$

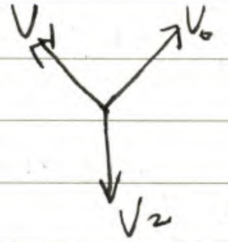
$\begin{pmatrix} 0 & 1 \\ -1 & a_d \end{pmatrix} \begin{pmatrix} V_{d-1} \\ \underset{V_0}{V_d} \end{pmatrix} = \begin{pmatrix} V_d \\ V_1 \end{pmatrix}$ $V_1 + V_{d-1} = a_d V_d$

- If $d = 3$ $\begin{pmatrix} 0 & 1 \\ -1 & a_3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & a_2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & a_1 \end{pmatrix} \begin{pmatrix} V_0 \\ V_1 \end{pmatrix} = \begin{pmatrix} V_0 \\ V_1 \end{pmatrix}$

$\Rightarrow a_1 = a_2 = a_3 = -1$

$\Rightarrow V_0 + V_2 + V_1 = 0$

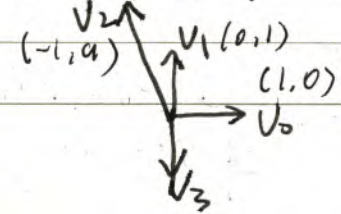
$\Rightarrow X(\Delta) = \mathbb{P}^2$



- If $d = 4$ $a_1 = a_3 = 0$ $a_2 = -a_4 \neq 0$

or $a_2 = a_4 = 0$ $a_1 = -a_3 \neq 0$

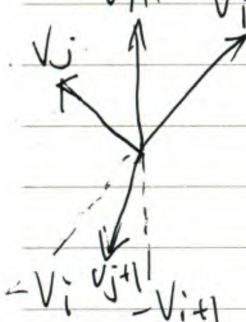
$a_1 = a$



$\Rightarrow X(\Delta) = F_a$ $\begin{cases} aV_1 = V_0 + V_2 \\ V_1 + V_3 = 0 \\ -aV_3 = V_2 + V_4 \end{cases}$

Claim: If $d \geq 5$, there must be some j,
$1 \le j \le d$, s.t. ~~v_{j-1} and~~
$$v_j = v_{j-1} + v_{j+1}$$

① for $\forall i, j$, v_j and v_{j+1} cannot be ~~are~~ arranged with
$$v_j \in \text{Relint}\left(\text{cone}(v_{i+1}, -v_i)\right)$$
and $v_{j+1} \in \text{Relint}\left(\text{cone}(-v_i, -v_{i+1})\right)$



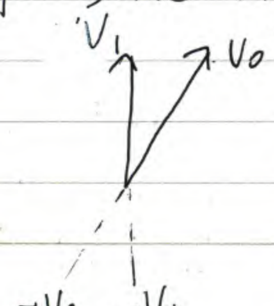
pf: Suppose
$$\begin{pmatrix} v_j \\ v_{j+1} \end{pmatrix} = \begin{pmatrix} -a & b \\ -c & -d \end{pmatrix} \begin{pmatrix} v_i \\ v_{i+1} \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 1 \\ -1 & a_j \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & a_{j-1} \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ -1 & a_{i+1} \end{pmatrix} \begin{pmatrix} v_i \\ v_{i+1} \end{pmatrix}$$

where $a, b, c, d \in \mathbb{Z}_{>0}$

But $\det \begin{pmatrix} -a & b \\ -c & -d \end{pmatrix}$
$$= ad + cb = 1$$
contradict!

② If $d \geq 4$, there must exist $v_j = -v_i$ for some i, j.

pf:



Consider $v_i \in \text{Relint}\left(\text{cone}(v_1, -v_1)\right)$

If $v_i \in \text{Relint}\left(\text{cone}(v_1, -v_0)\right)$
$\Rightarrow v_{i+1} = -v_0$

If $v_i \in \text{Relint}\left(\text{cone}(-v_0, -v_1)\right)$
$\Rightarrow v_{i+1} = -v_1$ If $v_i = -v_0$ done!

③ Suppose $V_i=-V_0$, $i\geq 3$, then there exists $0<j<i$ s.t $V_j=V_{j-1}+V_{j+1}$

pf: Let $V_j=-b_jV_0+b_j'v_1$ $(1\leq j\leq i)$

$(b_j, b_j'\geq 0,\ b_j+b_j'\neq 0)$

Let $c_j=b_j+b_j'$ $\cancel{c_j>0}$ $c_j\in\mathbb{Z}_{>0}$

$c_1=1$ $c_i=1$ $c_2\geq 2$

$\exists j$ s.t $c_j\geq c_{j-1}$, $c_j>c_{j+1}$

Then $a_jv_j=V_{j-1}+V_{j+1}$

$\Rightarrow (b_{j+1}'+b_{j-1}'-a_jb_j')v_1=(b_{j+1}+b_{j-1}-a_jb_j)v_0$

$\Rightarrow b_{j+1}'+b_{j-1}'-a_jb_j'$

$+b_{j+1}+b_{j-1}-a_jb_j=$

$=c_{j+1}+c_{j-1}-a_jc_j=0$

$\Rightarrow c_{j+1}-c_j+(c_{j-1}-c_j)=(a_j-2)c_j$

$\Rightarrow a_j-2<0 \Rightarrow a_j=1$

prop: All complete nonsingular toric surface are obtaind from $\mathbb{P}^2$ or F_a by a succession of blow-ups at T_N-fixed points